STATS 200A Discussion 7

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Theorem 7.1 (Central Limit Theorem)

$$X_1, X_2, \ldots, \overset{iid}{\sim} \mu$$

$\mathbb{E}(X_i), \mathbb{E}(X_i^2) < \infty$, Define

$$\bar{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$$

. $\mathbb{E}(X_i) := \mu, \mathrm{Var}(X_i) := \sigma^2$, then CLT:

$$\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \xrightarrow{d} N(0,1)$$

Proof.

$$X_Y : M_X = M_Y \Leftrightarrow X \overset{d}{=} Y$$

$$\begin{aligned}
\phi(t) &= M_{X_i}(t) \\
M_X(0) &= 1 = \mathbb{E}(e^0) \\
f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(t)\cdot h^n}{n!} \\
e^x &= e^0 + (e^0)'\cdot x + \frac{1}{2}(e^0)'' x^2 + \cdots \\
&= \sum \frac{x^n}{n!} \\
Z &= \sqrt{n}\cdot\frac{\bar{X}_n - \mu}{\sigma} \\
M_Z(t) &= \mathbb{E}(e^t Z) \\
&= \mathbb{E}\left(\exp\left(\frac{\sqrt{n}\bar{X}_n}{\sigma} - \frac{\sqrt{n}\mu}{\sigma}\right)\right) \\
&= \frac{\mathbb{E}\left(\exp\left(\frac{\sqrt{n}\cdot\bar{X}_n}{\sigma}\right)\right)}{\underbrace{\exp\left(\frac{\sqrt{n}\mu}{\sigma}\right)}_{\psi}} \\
&= \frac{1}{\psi}\cdot\mathbb{E}\left(\exp\left(\frac{\sqrt{n}}{\sigma}\frac{1}{n}\sum_{i=1}^{n} X_i\right)\right) \\
&= \frac{1}{\psi}\cdot\mathbb{E}\left(\prod_{i=1}^{n}\exp\left(\frac{1}{\sigma\sqrt{n}}X_i\right)\right)
\end{aligned}$$

$$
\begin{aligned}
(\text{iid}) &= \frac{1}{\psi} \cdot \prod_{i=1}^{n} \mathbb{E}\left(\exp\left(\frac{1}{\sigma\sqrt{n}} X_i\right)\right) \\
&= \frac{1}{\psi}\left(\phi\left(\frac{1}{\sigma\sqrt{n}}\right)\right)^n \\
&= \frac{1}{\psi}\left(\mathbb{E}\left(\exp\left(\frac{1}{\sigma\sqrt{n}} X_i\right)\right)\right)^n \\
&= \frac{1}{\psi}\left(\mathbb{E}\left(1 + \frac{1}{\sigma\sqrt{n}} X_i + \frac{1}{\sigma^2 n} X_i^2 + \cdots\right)\right)^n \\
&= \frac{1}{\psi}\left(1 + \frac{\mu}{\sigma\sqrt{n}} + \frac{\sigma^2 + \mu^2}{\sigma^2 n} + \cdots\right)^n \\
&= \lim_{n\to\infty} \frac{1}{\psi}\left(1 + \frac{1}{\sqrt{n}}\left(\frac{\mu}{\sigma} + \frac{\sigma^2 + \mu^2}{\sigma^2\sqrt{n}}\right)^{\sqrt{n}}\right)^{\sqrt{n}} \\
&= \frac{1}{\psi} \exp\left(\frac{\mu}{\sigma} + \frac{\sigma^2 + \mu^2}{\sigma^2\sqrt{n}}\sqrt{n}\right) \\
&= \frac{1}{\exp\left(\frac{\sqrt{n}\mu}{\sigma}\right)} \exp\left[\left(\frac{\mu}{\sigma} + \frac{\sigma^2 + \mu^2}{\sigma^2\sqrt{n}}\right)\sqrt{n}\right] \\
&= \exp\left(1 + \frac{\mu^2}{\sigma^2}\right)
\end{aligned}
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Definition 7.2 (Moment Generating Function)

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\begin{aligned}
M_{X_i}(t) &= \mathbb{E}(e^{t(X_i)}) \\
\left.\frac{\partial^{(r)} M_{X_i}(t)}{\partial t^{(r)}}\right|_{t=0} &= \mathbb{E}(X_i^r)
\end{aligned}
$$

$$
\mathbb{E}\left[e^{tX_i}\right] = \mathbb{E}\left[1 + t \cdot X + \frac{t^2}{2} X^2 + \frac{t^3}{6} X^3 + \cdots\right]
$$

$M_X = M_Y, \mathbb{E}(X^r) = \mathbb{E}(Y^r), X \overset{d}{=} Y$
$P(z) = \alpha_1 z + \alpha_2 z^2 + \alpha_3 z^3 + \cdots$
$\mathbb{E}(P(X)) = \mathbb{E}(P(Y))$
$C_b : \{f : \text{ continuous and bounded}\}, \{P(z)\}, \text{ dense } \in C_b \Rightarrow \mathbb{E}[f(X)] = \mathbb{E}[f(Y)], f \in C_b \Rightarrow X \overset{d}{=} Y$

Exercise 7.3

$$
N_1, N_2, \ldots, \overset{iid}{\sim} \text{Pois}(\alpha), \mathbb{P}(N_i = k) = \frac{\alpha^k e^{-\alpha}}{k!}
$$

a) Find MGF of N_i

b) Show $X, Y \overset{iid}{\sim} \text{Pois}(\alpha), X + y \sim \text{Pois}(2\alpha)$

c) $Z_n := \frac{\frac{1}{n} S_n - n\alpha}{\sqrt{\alpha}}$

Proof.

a)

$$\begin{aligned} M_{N_i} &= \mathbb{E}\left(\exp(tN_i)\right) \\ &= \sum_{k=0}^{\infty} \exp(tk) \cdot \frac{\alpha^k e^{-\alpha}}{k!} \\ &= e^{-\alpha} \sum_{k=0}^{\infty} \frac{(e^t\alpha)^k}{k!} \\ &= e^{-\alpha} \cdot e^{e^t\alpha} \end{aligned}$$

b)

$$\begin{aligned} M_{X+Y} &= \mathbb{E}(\exp(etX + tY)) \\ &= \mathbb{E}(e^{tX}e^{tY}) \\ \text{iid} &= \mathbb{E}(\exp(tX))\mathbb{E}(\exp(tY)) \\ &= e^{-\alpha} \cdot e^{e^t\alpha} \cdot e^{-\alpha} \cdot e^{e^t\alpha} \\ &= e^{-2\alpha} \cdot e^{e^t 2\alpha} \end{aligned}$$

$$S_n = \sum_{i=1}^{n} N_i S_n \sim \text{Pois}(n\alpha)$$

c)

$$\begin{aligned} \lim_{n\to Z_n} &\stackrel{?}{=} \\ M_{Z_n} &= \mathbb{E}\left(\exp\left(t\frac{\frac{1}{n}S_n - \alpha}{\sqrt{\alpha}}\right)\right) \\ &= \exp(t \cdot \frac{-\alpha}{\sqrt{\alpha}}) \cdot \mathbb{E}\left(\exp\left(\frac{\frac{t}{n}S_n}{\sqrt{\alpha}}\right)\right) \end{aligned}$$

Let $\tau = \frac{t}{n\sqrt{\alpha}}$

$$\begin{aligned} &= \exp(-t\sqrt{\alpha}) \cdot \mathbb{E}\left(\exp(\tau \cdot S_n)\right) \\ M_{S_n}(\tau) &= e^{-n\alpha}e^{e^\tau \cdot n\alpha} \\ Z_n &\sim N(0,1) \end{aligned}$$

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